Lecture plan · for the lecturer

Transactions and distributed data

Data engineering · University · Lecture 3

Time	Materials
60 min	Projector or board; The lecture slides and the seminar sheet from the kit; Phones or laptops with Darsi, for the lab

Learning goals

- State the ACID guarantees.
- Recognise isolation anomalies.
- Argue CAP trade-offs.

Activity: Interleave and fix

1. Warm-up (5 min): Show both confirmation emails and ask pairs to explain the bug.
2. Act (15 min): Two volunteers play T1 and T2 while the class calls out interleavings.
3. Reproduce (20 min): Pairs reproduce the lost update on a real database, then fix it.
4. Argue (15 min): Decide what registration and the library catalogue should do during a partition.
5. Wrap-up (5 min): Students write the muddiest point of the lecture and one question; read them before next week.

Discussion

- Which interleavings were safe?
- What is your default isolation level?
- Which system chose availability?

Exit ticket

- Name one operation that must stay consistent during a partition.

Try it in your next lecture

Give two volunteers cards with the steps of two transactions; the class calls out an interleaving and records the final seat count, then add a lock and try again.

Seminar sheet · for students

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Name _____ Group _____ Date _____

1 **During a partition, a ledger must never show two balances. What does it give up?**

◯ Availability ◯ Durability ◯ Consistency

2 **Count the outcome.**

seats = 1. T1 and T2 both read 1, both write seats - 1 and enrol a student. Final seats = _____ and students enrolled = _____.

Word bank: 1 · 0 · 2 · -1

3 **Which prevent the double booking? Choose all that apply.**

☐ Serializable isolation with retry ☐ Reading the count twice

☐ UPDATE course SET seats = seats - 1 WHERE id = 'CS450' AND seats > 0, checking rows affected

☐ SELECT … FOR UPDATE before writing

4 **Snapshot isolation can let two concurrent transactions end in a result that no serial order of them could produce.**

◯ True ◯ False

5 **Match each anomaly to the weakest level that prevents it in the ANSI model.**

Dirty read	• • • •	Serializable
Non-repeatable read	• • • •	Read committed
Phantom	• • • •	Repeatable read

Answers · for the lecturer

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1 **Availability**

It refuses some requests until the partition heals.

2 **seats = 1. T1 and T2 both read 1, both write seats − 1 and enrol a student. Final seats = 0 and students enrolled = 2.**

Each overwrote the other's write: a lost update that hides the overbooking.

3 **UPDATE course SET seats = seats - 1 WHERE id = 'CS450' AND seats > 0, checking rows affected; Serializable isolation with retry; SELECT … FOR UPDATE before writing**

Reading twice does not stop the other transaction from interleaving.

4 **True**

That is write skew (Berenson et al., 1995): snapshot isolation does not guarantee serializability.

5 **Dirty read — Read committed**
Non-repeatable read — Repeatable read
Phantom — Serializable

Each level adds one guarantee: read committed stops dirty reads, repeatable read also stops non-repeatable reads, and only serializable also stops phantoms.

✎ **For you to think about**

Which isolation level does the database your students use default to, and have they ever seen it fail?